George Kopf V

Electrical & Computer Engineering | Princeton University
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Education

Princeton University, Princeton, NJ — May 2027 (Expected)
B.S.E. Electrical & Computer Engineering, Minor in Robotics – 3.6 GPA

Relevant Coursework

- Intelligent Robotics, Power Electronics, Spaceflight Projects, Advanced Wireless Systems, Mechanical Design, Fluid Dynamics, Molecular Biology, Chemistry I & II. Upcoming: Differential Equations, Circuits & ICs

Technical Skills

- **Hardware:** Altium, KiCad, EasyEDA, LTspice, PLECS, SolidWorks, NX, Fusion, Creo
- **Software:** Python, C/C++, Java, Arduino, ROS2, LabVIEW, MATLAB, Verilog
- **Instrumentation:** Oscilloscopes, DMMs, AFGs, spectrum analyzers, PLCs, thermal-vacuum & vibration testing
- **Fabrication:** PCB soldering/reflow, lathe, mill, CNC, FDM/SLA printing
- **Wet Lab:** PCR, Gel Electrophoresis, SDS-Page, Bacterial & Mammalian Cell Culturing, Transformation

Engineering Experience

Robotics Research — Princeton University BAM Lab REU — Feb 2026 – Present

- Designed and fabricated an untethered biomimetic flying fish robot achieving 11 cm/s swimming at 2 Hz, with 2.4 GHz wireless telemetry; developing pectoral fin deployment and PID feedback control.
- Designed compact PCB integrating ESP32-S3, TPS61023 boost converter, and DRV8212 motor driver; developed fin mechanisms and waterproof chassis with sealed airbox and flooded wetbox. Maintained wireless telemetry using a surface-breaching antenna.
- Diagnosed motor-induced voltage transients affecting ESP32 operation – redesigned power distribution and decoupling circuitry to improve system reliability during swimming.

Robotics Research — Kent State University REU — Jun 2025 – Aug 2025

- Characterized and debugged an adjoint-based gradient descent optimization algorithm used to simulate and optimize fish swimming gaits.
- Developed LabVIEW control software for an experimental apparatus studying water flow in deformable tubing.

Bioengineering Research — Princeton University (iGEM) — Jun 2024 – Aug 2024

- Engineered novel probiotic bacteria therapy for ulcerative colitis treatment. Cultured mammalian cells for interaction tests with engineered cultures.
- Designed and fabricated microfluidic test chambers for biological response measurement. Project awarded 2024 iGEM Gold Medal.

Projects & Extracurriculars

Princeton Racing Electric (FSAE & F+HE) — Sep 2024 – Present

- Led water-cooling subteam, assisted 600 V safety architecture, and fabricated carbon-fiber aero with vacuum resin infusion.

Princeton University Robotics Club — Sep 2024 – May 2026

- Designed 9.6 V power architecture and eye/neck actuation for WALL-E; developed 6-legged RPi5-based hexapod.

TigerCub: PocketQube Satellite

- Designed, built, and tested a custom burn-wire deployment PCB for a deployable retroreflector on a 1P PocketQube satellite, switching 1 A at 5 V for 5 seconds.

Self-Driving Car

- Developed dual-layer iLQR safety filter for autonomous RC vehicle using Python, MPC, and ROS 2.

Search and Rescue Robot

- Led electrical integration for 11-person team; designed autonomous robot wiring harness distributing 12 V/9.6 V to 12 components.